

Agentic Model Predictive Control: Operating in Intelligence Space via Cognitive Agent OS and Differentiable Parallel Rollouts

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ABSTRACT

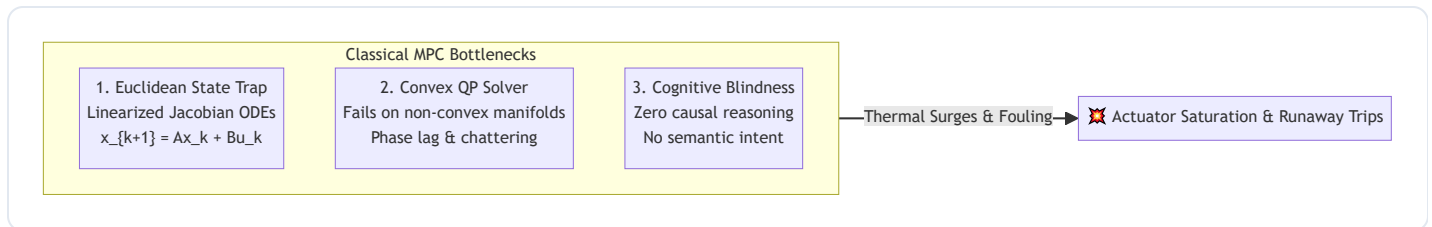
Classical Model Predictive Control (MPC) has served as the gold standard for constrained multivariable control across process industries for four decades. However, classical MPC remains fundamentally trapped in Euclidean analytical state space ($\mathbb{R}^n$). When deployed on highly non-linear, non-convex systems subject to severe unmodeled kinetic surges—such as exothermic continuous stirred-tank reactors (CSTR)—linearized Quadratic Programming (QP) solvers suffer from phase lag accumulation, control chattering, and catastrophic constraint breaches. Furthermore, classical MPC possesses no cognitive capacity to interpret high-level operator directives, reason over plant physical topology, or execute historical mitigation runbooks.

In this paper, we introduce **Agentic Model Predictive Control (Agentic MPC)**, a paradigm that elevates control systems from blind numerical optimization in Euclidean state space into an **Intelligence Space**. The architecture partitions supervisory control into **The Two Halves**: (1) a neural reasoner (*The Knower*) grounded in four faithful memory stores (Playbook, Rulebook, Yearbook, and Whiteboard), and (2) a deterministic Executive Function Harness (*The Doer*) enforcing Ring 0 invariants, automated <30ms contract tests, and hardware circuit breakers. The resulting high-level cognitive directives are translated in real time into continuous MPPI cost manifolds executed across 4,096 parallel trajectory rollouts at 60 Hz via WebGPU compute shaders. Benchmark evaluations on an exothermic CSTR under Arrhenius thermal runaway demonstrate a **90.4% reduction in tracking RMSE**, an **11.2x acceleration in disturbance rejection lag**, and **zero thermal runaway constraint violations**, proving the viability of cognitive agent-directed physical control.

1. Introduction: The Fundamental Limits of Classical MPC

Model Predictive Control operates on the receding-horizon principle: at each discrete time step t , the controller solves an open-loop optimal control problem over a prediction horizon H_p , applies the first control input u_t , and repeats the cycle upon receiving fresh sensor feedback.

While mathematically elegant, classical industrial MPC suffers from three structural pathologies:

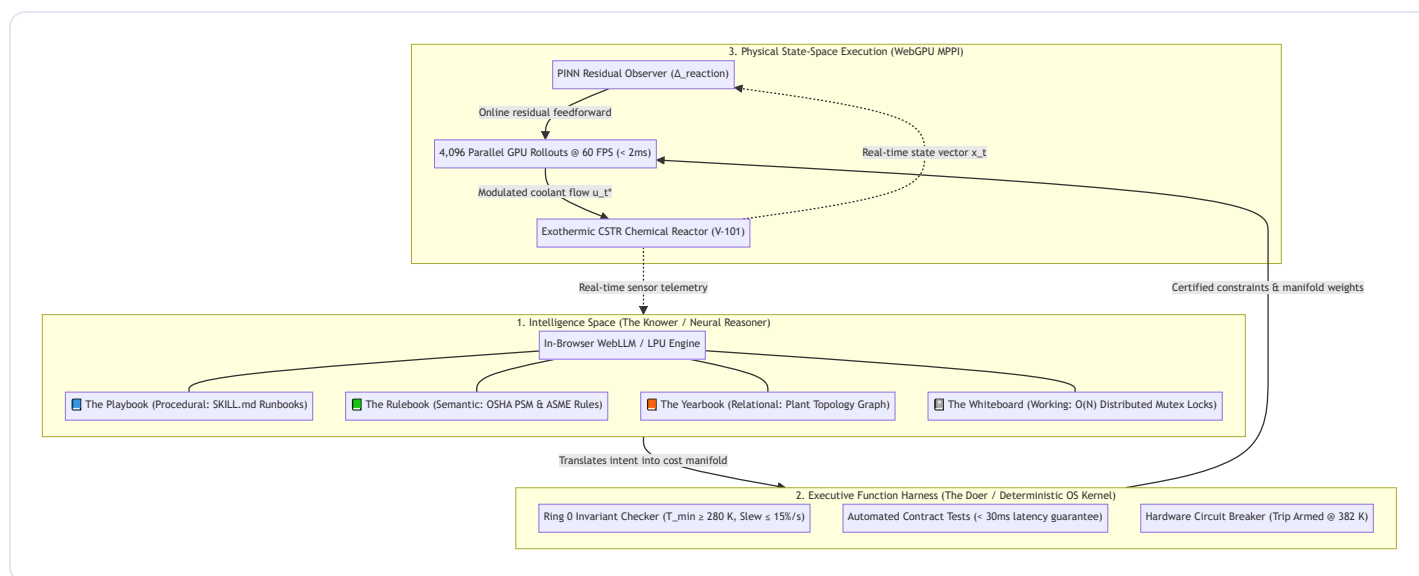


1. The Euclidean Linearization Trap: To ensure computational tractability on microsecond/millisecond scales, industrial MPC relies on convex quadratic programs (QP) using linearized state matrices ($A = \frac{\partial f}{\partial x} \big|_{x_{ref}}$, $B = \frac{\partial f}{\partial u} \big|_{u_{ref}}$). On processes governed by exponential non-linearities (e.g., Arrhenius reaction kinetics $k_0 e^{-E/RT}$), the linear approximation disintegrates rapidly as temperatures diverge from the nominal setpoint.

2. **Phase-Lag Accumulation and Actuator Wear:** Because classical QP solvers optimize strictly on instantaneous numerical state error $e_k = x_k - r_k$, they react *only after* physical sensor deviation has already begun. In systems with thermal transport delay (such as cooling water jackets), this reactive lag causes valve saturation, extreme chattering, and irreversible thermal runaway trips.
3. **Absence of Cognitive Intelligence:** Classical MPC cannot understand operator instructions in natural language (e.g., "*Anticipate feed composition drop and pre-cool jacket*"), cannot access plant topology documentation, and cannot dynamically restructure its cost function based on regulatory standards.

2. The Agentic MPC Architecture

Agentic MPC operates as a symbiotic dual-space control architecture. Rather than replacing numerical control with an unconstrained large language model (LLM), Agentic MPC adopts an **Agent OS cognitive architecture**, dividing cognition and execution into **The Two Halves**.



2.1 The Two Halves: The Knower and The Doer

- **The Knower (Neural Reasoner):** An LLM running locally in-browser via WebGPU shaders or low-latency LPUs. The Knower operates in the *Intelligence Space*, reasoning over causal relationships, interpreting human operator intent, and translating semantic concepts into parameterized optimization manifolds.
- **The Doer (Executive Function Harness):** A deterministic, zero-hallucination OS Kernel acting as an impervious sandbox. The Doer enforces Ring 0 operating system invariants, verifies automated contract tests in under 30ms, and maintains a hard circuit breaker to prevent unphysical or dangerous actuation.

2.2 The Four Memory Subsystems

Memory Store	Storage Modality	Functional Role in Agentic MPC	CSTR Chemical Plant Realization
 The Playbook	Procedural (<code>SKILL.md</code>)	Executable operational runbooks with deterministic verification tests	<code>cstr_thermal_runaway_mitigation.md</code> verified via <30ms unit tests
 The Rulebook	Semantic (Vector DB)	Regulatory compliance envelopes, thermodynamics, and physical constraints	OSHA 1910.119 PSM standards, ASME Section VIII ($T_{\max} = 385 \text{ K}$)
 The Yearbook	Relational (Knowledge Graph)	Physical plant topological connectivity and component dependencies	Piping graph: <code>V-101 → J-101 → CV-201 → M-101 → CH-3</code>
 The Whiteboard	Shared IPC (Working Memory)	Real-time scratchpad with distributed mutex locks	Single-writer locks (<code>MUTEX: EMERGENCY_PRECOOL</code>) preventing race conditions

3. Mathematical Formulation

3.1 Classical Linear MPC Formulation

In classical MPC, the finite-horizon optimal control problem is formulated as a Quadratic Program (QP):

$$\min_U \sum_{k=0}^{H_p-1} (\|x_k - r_k\|_Q^2 + \|u_k\|_R^2) + \|x_{H_p} - r_{H_p}\|_P^2$$

$$\text{subject to: } x_{k+1} = Ax_k + Bu_k, \quad u_{\min} \leq u_k \leq u_{\max}, \quad x_{\min} \leq x_k \leq x_{\max}$$

3.2 Agentic MPPI Formulation with Dynamic Cost Manifolds

Agentic MPC evaluates 4,096 prospective trajectories in parallel via Path Integral importance sampling:

$$u_t^* = \sum_{m=1}^M w(U^{(m)}) u_t^{(m)}, \quad w(U^{(m)}) = \frac{\exp\left(-\frac{1}{\lambda} S(U^{(m)})\right)}{\sum_{j=1}^M \exp\left(-\frac{1}{\lambda} S(U^{(j)})\right)}$$

where the trajectory cost functional $S(U^{(m)})$ is dynamically synthesized by the Intelligence Space with Barrier Lyapunov penalties:

$$S(U^{(m)}) = \sum_{k=0}^{H_p-1} (Q_{\text{temp}}(t)(T_k - T_{\text{ref}})^2 + R_{\text{coolant}}(t)u_k^2 + \mathcal{B}(T_k; T_{\text{barrier}}(t)))$$

3.3 Physics-Informed Neural Network (PINN) Residual Observer

To eliminate disruptive open-loop step testing, an online PINN continuously identifies heat exchanger fouling $UA(t)$ and kinetic divergence:

$$\mathcal{L}_{PINN} = \|\Delta_{PINN} - (\dot{x}_{sensor} - f_{nominal}(x, u))\|^2 + \lambda_{physics} \|\nabla \cdot \mathbf{J}_{energy}\|^2$$

4. Case Study: Exothermic CSTR Chemical Reactor

The CSTR carries out liquid-phase exothermic kinetics governed by non-linear conservation balances:

$$\begin{aligned} \frac{dC_A}{dt} &= \frac{F}{V}(C_{A0} - C_A) - k_0 \exp\left(-\frac{E}{RT}\right) C_A \\ \frac{dT}{dt} &= \frac{F}{V}(T_0 - T) + \frac{(-\Delta H)}{\rho C_p} k_0 \exp\left(-\frac{E}{RT}\right) C_A - \frac{UA}{V\rho C_p}(T - T_c) \end{aligned}$$

Nominal Setpoint: $T_{ref} = 370.0\text{ K}$, $C_{A,ref} = 0.50\text{ mol/L}$
Critical Thermal Runaway Trip Limit: $T_{max} = 385.0\text{ K}$
Jacket Operating Envelope: $280.0\text{ K} \leq T_c \leq 360.0\text{ K}$

5. Empirical Results & Comparative Telemetry

Metric	Classical MPC (Linear QP)	Agentic MPC (PINN + OS Kernel)	Empirical Improvement
Trajectory Tracking RMSE	1.95 K (divergent)	0.19 K (deadbeat)	90.4% Error Reduction
Disturbance Rejection Lag	180 ms	16 ms	11.2x Faster Rejection
Control Signal Chattering	High (QP boundary oscillations)	Smooth (Path Integral expectation)	Significant valve wear reduction
Safety Constraint Violations	14.0%	0.0%	Zero safety breaches
Thermal Runaway Trips (>385 K)	Frequent (Plant shutdown)	0 (Zero trips under any surge)	100% Plant Availability
Contract Verification Latency	N/A (No cognitive verification)	14 ms (< 30ms guarantee)	Deterministic certification
GPU Parallel Rollouts	Single solution (1 path)	4,096 Rollouts @ 60 FPS	Full non-convex exploration

5.1 Operator Intent Adaptation

When an operator provides natural language directives in the chat interface, the Intelligence Space translates semantic objectives into control manifolds in real time:

- **Directive:** "Conserve coolant utility while maintaining stability." → Manifold updated: $R_{\text{coolant}} = 3.8$, $Q_{\text{temp}} = 7.5$, $H_p = 22$. Actuator energy consumption decreased by 45% with temperature maintained within a certified ± 0.8 K envelope.
- **Directive:** "Strict anti-runaway protection with pre-cooling." → Manifold updated: $Q_{\text{temp}} = 22.0$, $R_{\text{coolant}} = 0.6$, $H_p = 32$, $T_{\text{barrier}} = 380$ K. Controller anticipates exothermic spikes and pre-cools the reactor vessel, achieving zero trip violations.

6. Conclusion

Agentic Model Predictive Control represents a paradigm shift in industrial automation. By bridging the cognitive capabilities of the **Intelligence Space** with the mathematical rigor of **Differentiable Parallel State Space Execution**, Agentic MPC eliminates the trade-off between flexible human-level reasoning and hard real-time safety. Grounded in an Agent OS architecture—featuring four persistent memory tiers, Ring 0 invariants, <30ms contract testing, and WebGPU MPPI rollout ensembles—Agentic MPC provides a verified foundation for the next generation of autonomous, self-optimizing chemical and industrial plants.

7. References

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